

Lecture 0: Introduction Notes.



Notation: x --- feature

b --- label

denote the i th data set as

$$[x^{(i)} \quad b^{(i)}]$$

$$x = [x_1, x_2, \dots, x_d]$$

where d is the dimension of the problem.

Task: Find a separating line in between.

① separate the 2 classes.

② leave as much "margin" as possible both classes.

Fact 1:

Given 2 parallel lines:

$$\begin{cases} a_1 x_1 + a_2 x_2 + x_3 = 0 \\ a_1 x_1 + a_2 x_2 + x_4 = 0 \end{cases}$$

$$a_1 x_1 + a_2 x_2 + x_4 = 0$$

here x_1, x_2, x_3

scalar

a_1, a_2 variables

the distance between them: $d = \frac{|x_4 - x_3|}{\sqrt{x_1^2 + x_2^2}}$

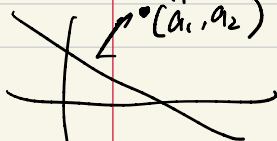
Fact 2: Given a line

$$a_1 x_1 + a_2 x_2 + x_3 = b$$

A point $(\hat{a}_1, \hat{a}_2)$ on the top side of the

line if $\hat{a}_1 x_1 + \hat{a}_2 x_2 + x_3 \geq b$

$p^*(\hat{a}_1, \hat{a}_2)$



suppose the classifiers we seek is

$$\textcircled{1} \quad \hat{x}_1 a_1 + \hat{x}_2 a_2 + \hat{x}_3 = 0, \quad \text{where } \hat{x}_1, \hat{x}_2, \hat{x}_3 \text{ are the parameter we optimize.}$$

Then the line that "supports" the "+" data

$$\textcircled{2} \quad \hat{x}_1 a_1 + \hat{x}_2 a_2 + \hat{x}_3 = C_1 \quad \text{"+"}$$

$$\textcircled{3} \quad \hat{x}_1 a_1 + \hat{x}_2 a_2 + \hat{x}_3 = -C_1 \quad \text{"-"}$$

normalize by C_1 (we can do this because it is the same line.)

That is, dividing $\textcircled{1}$, $\textcircled{2}$, $\textcircled{3}$, all by the constant C_1 , we obtain.

$$\frac{1}{C_1} (\hat{x}_1 a_1 + \hat{x}_2 a_2 + \hat{x}_3) = 0$$

$$\frac{1}{C_1} (\hat{x}_1 a_1 + \hat{x}_2 a_2 + \hat{x}_3) = 1$$

$$\frac{1}{C_1} (\hat{x}_1 a_1 + \hat{x}_2 a_2 + \hat{x}_3) = -1$$

re name $\frac{1}{C_1} \hat{x}_1$ as x_1 , $\frac{1}{C_1} \hat{x}_2$ as x_2

$\frac{1}{C_1} \hat{x}_3$ as x_3 , we arrive 3 equivalent lines

$$x_1 a_1 + x_2 a_2 + x_3 = 0$$

$$(x_1 a_1 + x_2 a_2 + x_3) = 1$$

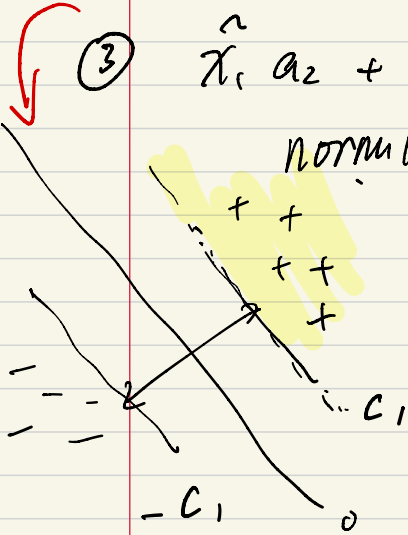
$$(x_1 a_1 + x_2 a_2 + x_3) = -1$$

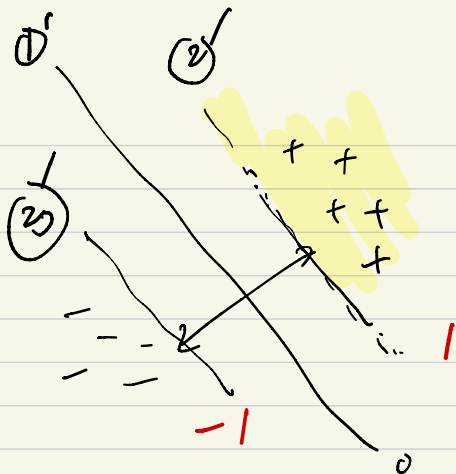
$\textcircled{1}$

$\textcircled{2}$

$\textcircled{3}$

Graphical illustration
next page





Goal: maximize the distance between lines ②' & ③', while ensuring line ②' supports the "+" data samples, line ③' supports "-" data samples.

optimization
formulation

$$b(i) \left(x_1 a_1^{(i)} + x_2 a_2^{(i)} + x_3 \right) \geq 1$$

$$\max_{x_1, x_2, x_3} \frac{2}{\sqrt{x_1^2 + x_2^2}}$$

(i)th data point

if $b(i) = +1$

$$x_1 a_1^{(i)} + x_2 a_2^{(i)} + x_3 \geq 1$$

if $b(i) = -1$

$$x_1 a_1^{(i)} + x_2 a_2^{(i)} + x_3 \leq -1$$

Note: $x_1 a_1^{(i)} + x_2 a_2^{(i)} + x_3 \geq 1$ ensures that the data sample $(a_1^{(i)}, a_2^{(i)})$ stays above the line $x_1 a_1 + x_2 a_2 + x_3 = 1$; See Fact 2.